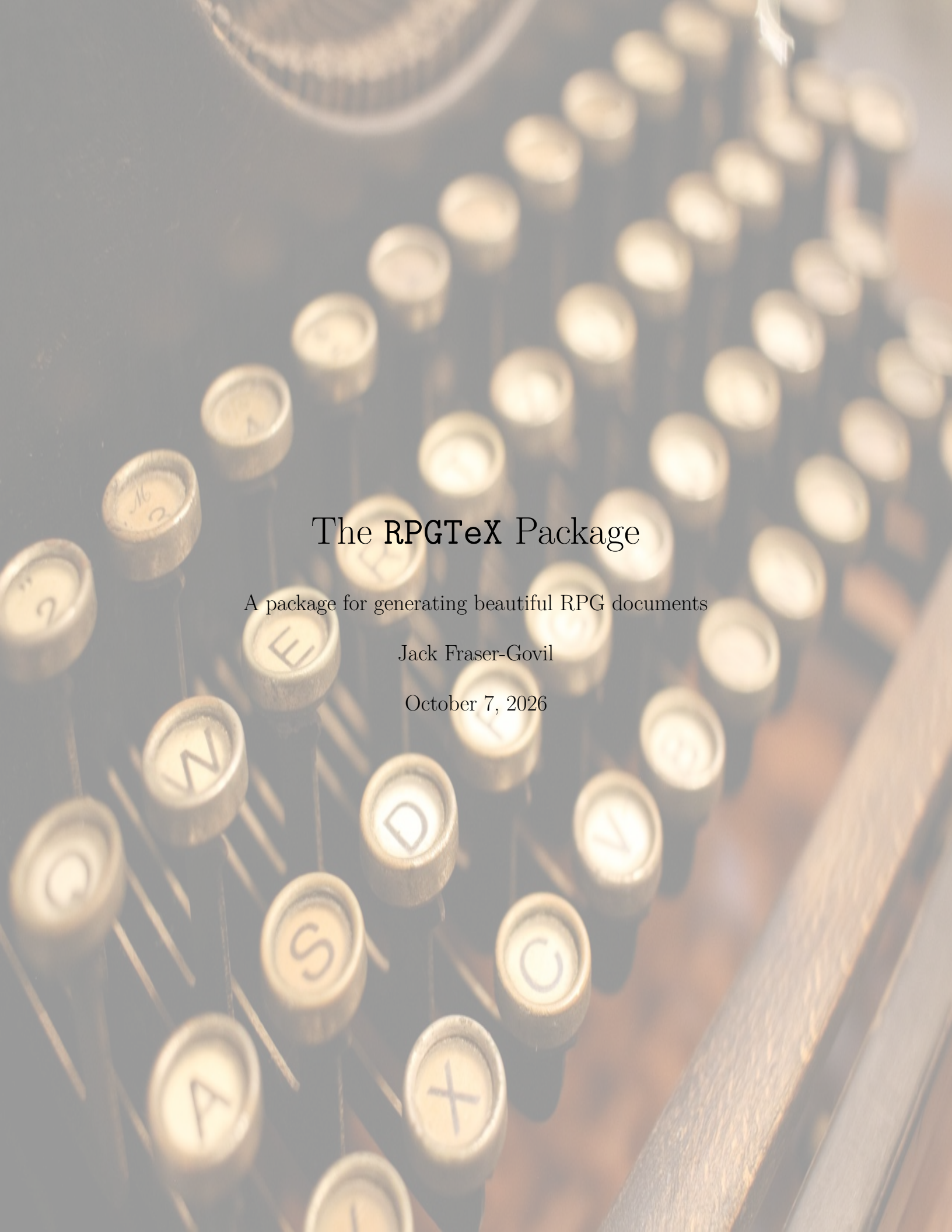




The **RPGTeX** Package

A package for generating beautiful RPG documents

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Welcome to the *RPGTeX* package. This $\LaTeX$ package is designed to allow users to flexibly typeset documents associated with Role Playing Games such as *Dungeons & Dragons* -- and many more besides. This package defines a central engine: `rpgcore` which define a number of useful functions and classes, and a flexible set of `themes` which control how those commands are rendered in the final document.

Getting Started

This document is the full reference manual. For a quick example of how to get started with your first *RPGTeX* document, see the [example documents](#), which demonstrate the basic features of this package in a readable fashion.

Attribution & License

This package would not have been possible without the team who developed [its predecessor, the 'DND 5e LaTeX Template'](#). That code was released under an MIT license, the text of which can be found in the `LICENSE` file. `rpgtex` is released under the same license.

AI Usage Declaration

The author used a handful of AI chat tools during development to help guide their thinking. The package itself was written entirely by hand: no AI tools were used to write any of its code. An AI assistant (Claude) helped edit this documentation. All mistakes and bad decisions are organic in nature.

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Chapter 1: Installing R_PG_TE_X

Getting R_PG_TE_X

R_PG_TE_X may be installed in several ways, described below. The package dependencies can be found in Appendix B (page 42). Once installed, R_PG_TE_X attempts to locate its internal resources (fonts and images) automatically. If this fails, manual configuration may be required, see page 2.

Operating Systems

The commands below are written assuming a Linux or macOS terminal. Windows users should consult the section on page 2 for more information.

Direct Installation

The simplest approach is to install R_PG_TE_X on the `texmf` path, where the compiler will find it automatically:

bash

```
mkdir -p "$(kpsewhich -var-value TEXMFHOME)/tex/latex" # make sure the directory exists
git clone https://github.com/DrFraserGovil/rpgtex.git "$(kpsewhich -var-value
    TEXMFHOME)/tex/latex/rpgtex"
```

This clones the repository into your personal L^AT_EX directory, which is typically `~/texmf` on Linux, `~/Library/texmf` on macOS, and `C:\Users\<username>\texmf` on Windows.

Indirect Installation

If you intend to modify R_PG_TE_X -- for example, by creating a new theme -- it is more convenient to keep it somewhere easily accessible. First, clone the repository into a location of your choice:

bash

```
git clone https://github.com/DrFraserGovil/rpgtex.git ~/path/to/rpgtex
```

The package must then be made visible to the compiler, in one of two ways.

Using TEXINPUTS

The `TEXINPUTS` environment variable adds directories to the compiler's search path. The `//` includes all subdirectories of the given directory, and the trailing colon keeps the standard search locations:

bash

```
export TEXINPUTS=~/path/to/rpgtex//:
```

The exact command depends on your shell; add it to your shell's configuration file to make it permanent.

Using a Symbolic Link

Alternatively, a symbolic link from the `texmf` directory to the repository allows the compiler to behave exactly as if R_PG_TE_X had been installed directly:

bash

```
mkdir -p "$(kpsewhich -var-value TEXMFHOME)/tex/latex"
ln -sf ~/path/to/rpgtex "$(kpsewhich -var-value TEXMFHOME)/tex/latex/rpgtex"
```

Windows

RPGTeX works on Windows, with a few differences from the instructions above:

- Run the git commands using Git Bash for Windows
- Use the direct installation. Symbolic links and `TEXINPUTS` behave differently on Windows, and are best avoided.
- If you use MiKTeX, it does not automatically detect new files in its directories. After installing, register the folder containing RPGTeX as a root directory in MiKTeX Console (if it is not already one), and then refresh the file name database (*Tasks* → *Refresh file name database*). TeX Live requires no extra steps.
- Run the provided Python scripts through Python explicitly, for example `python rpglatex document.tex`. The `rpglatex` compiler does not yet open the finished PDF automatically on Windows.

Overleaf (Not Recommended)

We do not recommend Overleaf: the compilation time allowed by its free tier is too short for documents which use complex packages such as this one. Nevertheless, RPGTeX can be used there as follows:

1. Download the RPGTeX repository from GitHub as a ZIP archive.
2. On Overleaf, create a new project using *Upload Project*, and upload the ZIP archive.

Compiling RPGTeX Documents

RPGTeX uses the `fontspec` package to load its fonts, and so must be compiled with `xelatex` or `lualatex` in almost all cases. An exception is if RPGTeX is loaded as a package without the font option (page 9):

```
xelatex main.tex # works
lualatex main.tex # works
pdflatex main.tex # fails when the font option is active
```

Path Configuration

In most cases, no further steps are required to compile a document, since RPGTeX finds its own installation path, and stores the value in the following variable:

`\RpgPackagePath` (*Variable*)

The path to the root directory of the RPGTeX installation. RPGTeX uses it to load resources, such as fonts, which TeX cannot find relative to the package itself. It is set by the first of the following which is available:

1. **The `rpglatex` compiler** (page 41), which defines it before the document is read.
2. **A configuration file**, `core/rpg-config.cfg`, within the installation
3. **Automatically**. RPGTeX asks `kpsewhich` where `rpgtex.sty` is installed. This uses the compiler's restricted shell escape, which TeX Live enables by default, so no extra flags are needed.

If none of these succeeds, compilation stops with an error describing these options.

Failed Configuration

Most users will rely on the third option, the automatic configuration. However, this may fail if L^AT_EX is (for example) running in safe mode. In such cases the user must either use the `rpgtex` compiler, run the python script located at `scripts/configure-rpgtex`, or manually create the configuration file:

```
cd <rpgtex root directory>
echo '\edef\RpgPackagePath{"$(pwd)"}' > core/rpg-config.cfg
```

If RPGTeX is later moved, the file must be deleted or recreated.

PART I

RPGTeX For Users

Chapter 2: Package & Class Usage

RPGTeX can be used either as a standalone package, or through one of its classes.

Class	Purpose
<code>rpgbook</code> (page 4)	Larger RPG documents, such as rulebooks and adventures. Based on the standard <code>book</code> class.
<code>rpgcard</code> (page 6)	Individual cards, such as items, spells or abilities, for digital use.
<code>rpgdeck</code> (page 7)	Collections of cards, arranged for printing.
<code>rpghandout</code> (page 5)	Shorter documents, such as player handouts. Based on the standard <code>article</code> class.

Loading a one of these classes automatically activates the `font`, `sections`, `title` and `background` options (page 8)

Standalone Package

The package may be loaded into a document of any class:

```
\documentclass{<any-class>}
\usepackage[<options>]{rpgtex}
```

This loads the core commands and environments, and the theme (the `default` theme, unless the `theme` option is used). However, the modules which change the fonts, page backgrounds, section headings and title page are inactive unless they are enabled with the `font`, `background`, `sections` and `title` options (page 8). This allows RPGTeX to be added to an existing document without changing its overall appearance.

rpgbook

The `rpgbook` class is designed for writing long form documents such as rulebooks and sourcebooks for RPGs - cases where you need to be able to organise things into parts and chapters!

Inherited Class

The `rpgbook` class inherits from [the `extbook` class](#). This is an extension to the basic `book` class to allow more font sizes to be accepted. Otherwise it behaves near-identically to the standard `book` class.

The full list of sizes which `extbook` can accept (and thus allowed inputs for the `size` option (page ??)) is ``eight, nine, ten, eleven, twelve, fourteen, seventeen and twenty points''.

Special Commands

`rpgbook` inherits the following notable commands from the `book` class, which are not available in other classes:

Geometry

The default geometry for an `rpgbook` is:

Element	Size
Left and right margin	0.65in
Top margin	0.4in
Bottom margin (from main text to page bottom)	0.75in
Bottom margin (from main text to top of footer area)	0.3in
Gap between columns in <code>twocolumn</code> mode	0.25in

Options & Interactions

The `rpgbook` interacts with all of the options detailed on page 8. Note that there is no 'forwarding' to the underlying class and that there is a slightly different syntax for, i.e., setting the global font size.

- RPG books use a cover-title by default (`\RpgSetTitleMode` (page ??) is set to `cover`)
- The `extbook` provides the `part` and `chapter`
- The `layout` mode is activated
 1. Unless print mode is active, the page background will use the image set by `\RpgSetPaper` (page ??)
 2. Calling `\RpgSetTheme` clears the page (so that the old theme may complete)
- A table of contents is available and formatted using the ToC-fonts

rpghandout

The `rpghandout` class is designed for smaller documents where the full structure of a book is unnecessary, such as printouts for players, or individual adventure modules.

Inherited Class

The `rpghandout` class inherits from [the `extarticle` class](#). This is an extension to the basic `article` class to allow more font sizes to be accepted. Otherwise it behaves near-identically to the standard article class.

The full list of sizes which `extarticle` can accept (and thus allowed inputs for the `size` option (page ??)) is 'eight, nine, ten, eleven, twelve, fourteen, seventeen and twenty points'.

Special Commands

Geometry

The default geometry for an `rpghandout` is:

Element	Size
Left and right margin	0.65in
Top margin	0.4in
Bottom margin (from main text to page bottom)	0.75in
Bottom margin (from main text to top of footer area)	0.3in
Gap between columns in <code>twocolumn</code> mode	0.25in

Options & Interactions

The `rpghandout` interacts with all of the options detailed on page 8. Note that there is no 'forwarding' to the underlying class and that there is a slightly different syntax for, i.e., setting the global font size.

- RPG books use a header-title by default (`\RpgSetTitleMode` (page ??) is set to `header`)
- The `layout` mode is activated
 1. Unless print mode is active, the page background will use the image set by `\RpgSetPaper` (page ??)
 2. Calling `\RpgSetTheme` clears the page (so that the old theme may complete)
- A table of contents is available and formatted using the ToC-fonts

rpgcard

The rpgcard class is designed for typesetting individual (or very few) **RpgCard** (page 25) objects, primarily for digital and screen viewing. Printing of RpgCards is better done via a deck (page 7).

Inherited Class

The rpgcard class changes which class it inherits from depending on the *background* option which is passed, which is used as a proxy for 'print mode'.

Normal Mode. By default, the rpgcard class inherits from the **standalone** class. The final output pdf is cropped to the smallest rectangle which contains the output. This makes it ideal for screen-viewing a single card.

Print Mode. If the option **bg=print** or **bg=none** has been passed, the class instead inherits from **article**. This prevents the cropping, and makes it simple to print the card out as a 'normal' size. However, we recommend using the **RpgDeck** class for assembling and printing large numbers of cards.

Special Commands

The rpgcard class does not define any special commands.

Geometry

An rpgcard has 'no geometry' in the normal sense; as a standalone environment it resizes to fit the contents. The document expands horizontally if a `\cardbreak` (page 24) occurs, or if more than one card is added into the document. There is no way to add a linebreak -- this is usually an indication that you want an rpgdeck.

When in print mode, the margins are a uniform 5pt to prevent printing errors.

Options & Interactions

The rpgcard has the following special interactions with the options detailed on page 8:

Option	Normal Behaviour	rpgcard Behaviour
bg	Determines if the 'background paper' and footer decorations are rendered.	Background paper and footers are disabled. Command instead toggles the Inherited class/page geometry of the class (see above).
columns	Sets the number of columns	Ignored

Interactions

- rpgcards cannot set titles or use tables of contents
- rpgcards are generally unsuited for any environment which spans an entire page width, unless wrapped inside another environment. **RpgTable** (page 20) will fail if used 'exposed' in an rpgcard document; but works within an **RpgCard**.
- All switches (`\RpgSwitch` (page 18)) are set to true, setting everything into card-mode.

rpgdeck

The `rpgdeck` class is designed to allow a writer to aggregate a large number of `RpgCard` (page 25) objects into a single document. In particular, it is designed to allow easy consolidation of cards which have been individually typeset using a `rpgcard` (page 6) document.

The `rpgdeck` class is largely identical to the `rpghandout` class, with the following exceptions:

1. The default geometry is a simple 5pt margin on all sides.
2. The ability to activate `twocolumn` mode is deactivated
3. The `clear` pagestyle is set globally; no footer decorations are rendered (only the `\RpgSetPaper` (page ??)).
4. All switches (`\RpgSwitch` (page 18)) are set to true, setting everything into card-mode.

Including rpgcards

The `rpgdeck` activates the *standalone package*, with the `subpreambles=true` option passed.

This allows a writer to `\input` a standalone document, even if it has its own `\documentclass` and `\begin{ document }`, which would normally prohibit such an inclusion.

Example Deck-Building

Consider the following `rpgcard` document:

```
% spell1.tex
\documentclass[theme=dnd]{ rpgcard }
\def\spellname{ <text> }
\begin{ document }
  \begin{ RpgSpell }{ \spellname }
    (...)
  \end{ RpgSpell }
\end{ document }
```

This generates a single playing-card size description of the spell, and can be compiled on its own and (for example) sent to a player digitally. We can assume that we have multiple such documents, `spell2.tex` and so on.

If we wished to include this card in a larger document:

```
% deck1.tex
\documentclass[theme=dnd]{ rpgdeck }
\input{ spell1 }
\input{ spell2 }
\input{ spell3 }
(...)
\end{ document }
```

Quirks

In the example above we defined the name of the environment through the macro `\spellname`, which was defined in the preamble. This works fine, because the package invokes `\RequirePackage[subpreambles=true]{ standalone }`; the 'subpreambles' are parsed and included, even though they were invoked outside the preamble of the deck-document.

However, the package does this by collecting all the subpreambles first (excluding the `documentclass` declaration), then adding them to the main preamble, and then including the body text. This means that repeated definitions will override each other; if we had used `\spellname` for `spell2` and `spell3`, then *all* of the objects in the final deck would act as if they had `spell3`'s name. This can be avoided by moving the `\def` inside the `\begin{ document }`.

Note too that since the theme (`dnd`) of the card was defined in the `documentclass`, this is excluded when it is aggregated; only the 'deck' package options are executed. It is therefore possible to aggregate objects into a deck which uses a different theme than the original items; the items will be rendered using the parent theme. If this is a concern, a manual call to `\RpgSetTheme` (page 12) can be used.

Chapter 3: Options & Configuration

This chapter describes the ways in which the behaviour of R_PG_TE_X may be configured without writing any new code: the options which may be passed to the package or class, the named colours and lengths used by its elements, and the command line interface for changing these options at compile time.

Package Options

Whether R_PG_TE_X is loaded directly as a package or indirectly through one of its classes, options may be passed to it which change the behaviour of the resulting document:

```
\documentclass[<options>]{rpgbook}  
\usepackage[<options>]{rpgtex}
```

Each option takes one of two forms:

Key-value options: The name of the option is followed by an equals sign and a value:

```
\usepackage[bg = print, theme = dnd]{rpgtex}
```

Flags: Only the name of the option is provided. Its presence enables the associated behaviour:

```
\usepackage[justified]{rpgtex}
```

The available options are:

background <flag>

If present, activates the ‘paper background’ modules, which control the paper colour and background images, the footer decorations, and the spacing around floats.

This flag is set by default when loading a class, but must be passed explicitly to enable these modules when loading the package directly.

bg = full / print / none

Controls the behaviour of the background and footer decorations. The value must be one of the three listed above, else an error is thrown. The default value is **full**.

The most visible effect of this option is to control whether the ‘background paper’ set by `\RpgThemePaperImage` (page 33) and `\RpgThemePaperColor` (page 33), and the footer decorations set by `\RpgThemeFooterDecoration` (page 33), appear in the document:

Value	Paper	Footer Decorations
full	✓	✓
print	✗	✓
none	✗	✗

This option also changes the behaviour of the `rpgcard` class (page 6), and other environments may similarly alter their colours or layouts in response to its value.

columns = 1 / 2

Places the document into either `onecolumn` or `twocolumn` mode. The default value is 2.

More advanced column layouts require the user to manually use the `multicols` environment.

This option only has an effect when using one of the `RPGTeX` classes.

`font` <flag>

If present, activates the font modules, applying the package fonts wherever possible.

This flag is set by default when loading a class, but must be passed explicitly to enable these modules when loading the package directly.

`justified` <flag>

If present, sets the text to be justified throughout the document. By default, text is left-aligned ('ragged right').

`noemph` <flag>

If present, the default behaviour of `\emph` is left unchanged. By default, `\emph` is redefined to be an alias of `\RpgKey` (page 16).

`nomultitoc` <flag>

If present, disables the multi-column table of contents.

`oneside` <flag>

If present, places the document into one-sided mode. This disables alternating footers, but allows for consistent asymmetric margins (as in this document, where the left margin is enlarged to accommodate macro names).

This option only has an effect when using one of the `RPGTeX` classes.

`openright` <flag>

If present, places the document into right-opening mode, such that chapters always begin on a right-hand page. By default, chapters may begin on any page (see the documentation of the `book` class for more details).

This option only has an effect when using one of the `RPGTeX` classes.

`sections` <flag>

If present, activates the sections modules, including the part-page overloads and any theme behaviour which modifies the section commands.

This flag is set by default when loading a class, but must be passed explicitly to enable these modules when loading the package directly.

`size` = <font-size>pt

Sets the base font size of the document, if permitted by the parent class. The default value is 10pt.

This option only has an effect when using one of the `RPGTeX` classes.

The allowed values depend on the class being used (see the class section, beginning on page 4).

`theme` = <theme name>

The initial value passed to `\RpgSetTheme` (page 12) once package initialisation is complete.

The default value is `default`, which activates the Default Theme (page ??).

`themepath = </path/to/themes/>`

Calls `\RpgSetThemePath` (page 12) before the theme is loaded, allowing custom themes to be used.

The default value is `themes`, which `TEX` locates within the `RPGTeX` installation. A theme called `name` is then expected to be stored in `<themepath>/name/name.cfg`.

`title <flag>`

If present, activates the titling module and any theme presets which alter the behaviour of `\maketitle`.

This flag is set by default when loading a class, but must be passed explicitly to enable this module when loading the package directly.

Colors

`RPGTeX` defines a number of named colors which are used by its various elements. The default values listed below may be overridden by the active theme.

A Note on Spelling

Although this document otherwise uses British spelling, this section follows the American spelling of ‘color’ used throughout `LATEX` and `xcolor`. The color names below must be written exactly as given, including their spelling and capitalization.

An exception is in Gray/Grey, as I personally cannot remember which is the ‘right’ way to spell it, so I have aliased most colours to use both spellings.

Co-varying Colors

The following colors are coupled together, and by default all take the same value:

Color	Usage	Default
<code>SidebarColor</code>	The background color of the <code>RpgSidebar</code> (page 19) environment.	<code>ThemeColor</code>
<code>TableColor</code>	The background color of alternate rows in an <code>RpgTable</code> (page 20).	<code>ThemeColor</code>
<code>ThemeColor</code>	A general-purpose color from which the other co-varying colors inherit.	<code>white!60!black</code>
<code>TipColor</code>	The background color of the <code>RpgTip</code> (page 19) environment.	<code>ThemeColor</code>

Calling `\RpgSetThemeColor` (page 12) assigns a new value to `ThemeColor` and all three co-varying colors simultaneously:

```
\RpgSetThemeColor{blue!20!white}
```

Independent Colors

The remaining colors are set individually:

Color	Usage	Default
<code>ContourInnerColor</code>	The default color of the inner text produced by <code>\RpgContour</code> (page 15).	<code>black</code>
<code>ContourOuterColor</code>	The default color of the outer contour drawn around text by <code>\RpgContour</code> (page 15).	<code>white!50!black</code>
<code>FiligreeBackColor</code>	The background color of the <code>RpgFiligree</code> (page 21) environment.	<code>white</code>
<code>FiligreeColor</code>	The color used to draw the filigree frame applied by the <code>rpgfiligree</code> style.	<code>black</code>
<code>NarrationColor</code>	The background color of the <code>RpgNarration</code> (page 19) environment.	<code>F7F2E5</code> (HTML)
<code>PaperColor</code>	The background color of each page, applied only when the paper background is enabled (see the <code>bg</code> option, page 8).	<code>white</code>
<code>SecretColor</code>	The default frame color of the <code>RpgSecret</code> (page 22) environment.	<code>black!30!white</code>

Any of these colors, including the co-varying colors, may be modified individually using the standard `xcolor` interface:

```
\definecolor{NarrationColor}{HTML}{FFFFFF}  
\colorlet{SidebarColor}{red!10!white}
```

Modifying a co-varying color in this way changes only that color, leaving the others unaffected.

Lengths

A small number of lengths used by R_PG_TE_X may be changed directly, without any special interface. As with the colors, the active theme may override their default values.

`\RpgDropCapSeparation` (Variable)

The horizontal space between the enlarged letter created by `\RpgDropCap` (page 16) and the text which follows it. It is a standard L^AT_EX length, and is changed with `\setlength`:

```
\setlength{\RpgDropCapSeparation}{0.5em}
```

The default value is the width of a single space.

`\RpgFiligreeSize` (Variable)

The size of the decorative elements of the filigree frame (page 21), which the theme uses to scale its design. Unlike a standard length, it is stored as a macro, and so is changed with `\renewcommand`:

```
\renewcommand{\RpgFiligreeSize}{8pt}
```

The default value is 5pt.

Command Line Interface

L^AT_EX does not natively provide a 'command line interface' through which a document may be modified at compile time: any change must be made within the source file itself. It is, however, often convenient to compile the same document in several configurations without editing it -- most notably to produce both a standard version and a printer-friendly version (using `bg = print`, which removes the paper background to reduce ink usage).

To support this, R_PG_TE_X checks for a pseudo command-line variable when processing its options. Because T_EX treats the text passed to it on the command line as the beginning of its input stream, this variable can be defined before the document itself is read.

`\RpgCMD` (Variable)

If defined, holds a comma-separated list of package options (page 8), which are applied after all other options have been processed. The options are applied in the following order, with later values overriding earlier ones:

1. The defaults set by the class (if one is being used).
2. The options passed in the document, via `\documentclass` or `\usepackage`.
3. The contents of `\RpgCMD`.

Since the options are processed before they are passed to the parent class, options which affect the class (such as `columns` or `size`) may also be set in this way.

`\RpgCMD` is defined by passing a short piece of T_EX code to the compiler before the document is input:

```
xelatex "\def\RpgCMD{<rpg-options>} \input{<document>}"
```

This compiles `<document>` as though `<rpg-options>` had been passed to the class or package. For example, the following produces a printer-friendly, single-column version of a document without modifying the source:

```
xelatex "\def\RpgCMD{bg=print, columns=1} \input{document.tex}"
```

The `rpglatex` compiler (page 41) performs this insertion automatically: the `-p` (`--print`) flag sets `bg=print`, allowing the user to switch between the standard and print versions of a document with a single flag:

```
rpglatex document.tex -p
```

Chapter 4: Commands & Macros

Themes & Colours

`\RpgSetTheme { name }`

Loads the theme called **name**, replacing the appearance of any previously loaded theme. The theme is first searched for at `<themepath>/<name>/<name>.cfg`; if no such file exists, **name** is instead treated as a direct path to a theme file. An error is raised if neither file exists.

The initial theme is set by the **theme** option (page 8). `\RpgSetTheme` may be called at any point in the document, but since its effects are immediate, it is best called at the start of a new page.

`\RpgSetThemeColor { colour }`

Sets **ThemeColor** and all of the co-varying colours to **colour** (page 10).

`\RpgSetThemePath { path }`

Sets the directory searched by `\RpgSetTheme`, allowing custom themes to be stored outside of the **RPGTeX** installation. The path is not checked until a theme is loaded. The initial value is set by the **themepath** option (page 10).

Document Structure

RPGTeX extends the standard **L^AT_EX** commands for structuring a document. It provides title and part pages designed by the active theme, records the names of the current headings so that they may be reused elsewhere, and allows ‘fake’ headings to be added to the table of contents.

The Title & Sections Options

Much of the behaviour described in this section is provided by redefining standard **L^AT_EX** commands, and is controlled by two package options (page 8). Both are active by default when using one of the **RPGTeX** classes; otherwise, they must be passed to the package explicitly.

title Redefines `\maketitle` to produce the title designed by the active theme (equivalent to calling `\RpgMakeTitle`). Otherwise, `\maketitle` retains its original **L^AT_EX** behaviour.

sections Activates *sections mode*, which:

- Redefines `\chaptermark`, `\sectionmark`, `\subsectionmark` and `\subsubsectionmark` so that, in addition to their usual behaviour, they store the name they are given (page 14).
- Sets `\secnumdepth` to 0, so that sections and their subdivisions are unnumbered.
- Replaces `\part` with a version which records the part name and draws a full-page part page using the active theme. The starred form, `\part*`, records the name but otherwise falls back to the original **L^AT_EX** command.

Themes may additionally alter the formatting of the sectioning commands when sections mode is active.

All of the other commands in this section, including `\RpgMakeTitle`, are available regardless of these options, although the stored heading names are only updated in sections mode.

Title Pages

The title page draws on the usual `\title` and `\author` commands, along with the following additions.

`\cover { path/to/image }`

Sets an image to be used as the full-page background of the cover page created by `\maketitle` in `cover` mode (see below). The theme's title design is drawn over the top of the image.

The default value is empty, in which case the title design is drawn on a plain page.

`\maketitle` `{}`

`\RpgMakeTitle` Creates the title of the document using the design provided by the theme (only when the `title` option is active; see above). There are two title modes:

cover A full-page cover, drawn over the `\cover` image (if one has been set). This is the default for the `rpgbook` class.

header A title heading placed at the top of the first page of text.

The mode is selected with `\RpgThemeTitleMode{ cover / header }`. If the theme does not define a title for the active mode, the original L^AT_EX `\maketitle` is used instead.

`\RpgMakeTitle` is equivalent to `\maketitle`, but remains available even if the `title` option is not active, or if another package redefines `\maketitle`.

`\RpgCoverTransparency` `{ value }`

Fades the `\cover` image towards white, which can help to keep the title legible over a busy image. The `value` is the opacity of a white layer placed over the image, between 0 (the image is unaltered) and 1 (the image is completely hidden). The default value is 0.

`\RpgGetAuthor` *(Variable)*

`\RpgGetAuthor*` The author set by `\author`, formatted with the `\RpgFontSubTitle` (page 31) font element. The starred form gives the author as plain text.

`\RpgGetSubtitle` *(Variable)*

`\RpgGetSubtitle*` The subtitle set by `\subtitle`, formatted with the `\RpgFontSubTitle` (page 31) font element. The starred form gives the subtitle as plain text.

`\RpgGetTitle` *(Variable)*

`\RpgGetTitle*` The title set by `\title`, formatted with the `\RpgFontTitle` (page 31) font element (page 30). The starred form gives the title as plain text, without applying the font.

`\subtitle` `{ text }`

Sets the subtitle of the document. As with `\title` and `\author`, this has no visible effect by itself: the subtitle is displayed by `\maketitle` if the theme includes it. The default value is empty.

Part Pages

`\part` *(*)* `[<image>]{ name }`

In sections mode (page 12), `\part` is redefined to create a full-page part page using the design provided by the theme. Its behaviour depends on the form used:

`\part{name}` Adds the part to the table of contents and creates a part page on a plain background, or over the default part image if one has been set with `\RpgDefaultPartImage`.

`\part[image]{name}` As above, but uses `image` as the full-page background.

`\part*{name}` Uses the original `\part` command of the underlying class. Any image argument is ignored.

In all cases, `name` is recorded in `\RpgPartName` (page 14).

`\RpgDefaultPartImage` `{ path/to/image }`

Sets an image to be used as the full-page background of every part page which is not given its own image. By default, no image is used.

`\RpgPartTransparency { value }`

Fades all subsequent images set by `\part` image towards white, which can help to keep the text legible over a busy image. The `value` is the opacity of a white layer placed over the image, between 0 (the image is unaltered) and 1 (the image is completely hidden). The default value is 0.

Section Names

Many RPG documents display the name of the current chapter or section outside of the heading itself -- in a running header, a footer, or a decorative page border, for example. To support this, `RPGTeX` records the names of the current part, chapter, section, subsection and subsubsection, so that they may be reused anywhere in the document.

These names are only recorded in sections mode (page 12).

Each of the following variables is initially blank, and holds its value until the next command of the same level is called: beginning a new chapter does not, for example, reset `\RpgSectionName`.

`\RpgChapterName` *(Variable)*

The name given to the last `\chapter` command, or set by calling `\chaptermark`. Starred chapters (`\chapter*`) do not update this value.

`\RpgPartName` *(Variable)*

The name given to the last `\part` or `\part*` command.

`\RpgSectionName` *(Variable)*

The name given to the last `\section` command, or set by calling `\sectionmark`. Starred sections (`\section*`) do not update this value.

`\RpgSubsectionName` *(Variable)*

The name given to the last `\subsection` command, or set by calling `\subsectionmark`. Starred subsections (`\subsection*`) do not update this value.

`\RpgSubsubsectionName` *(Variable)*

The name given to the last `\subsubsection` command, or set by calling `\subsubsectionmark`. Starred subsubsections (`\subsubsection*`) do not update this value.

`\RpgGetName { part / chapter / section / subsection / subsubsection }`

Returns the name stored for the given level (equivalent to the corresponding variable above). Any other input results in an error.

Since these names (other than that of the part) are taken from the `mark` commands, they may be set manually, without creating a new heading, by calling the relevant `mark` command directly.

```
We are in Chapter '\RpgGetName{chapter}',  
section '\RpgGetName{section}'.
```

```
\subsubsection{An example subsubsection}
```

```
Now I am in  
'\RpgGetName{subsubsection}'.
```

We are in Chapter `Commands & Macros',
section `Document Structure'.

An example subsubsection

Now I am in `An example subsubsection'.

Fake Sections

It can be helpful to be able to integrate common RPG elements (such as full page maps or images) into the document structure. The following command provides a simple interface for spoofing in a section (or part, chapter etc.) into the table of contents, without actually invoking a sectioning command:

```
\RpgFake { level }{ name }
```

Mimics the creation of a new heading at the given `level` without printing a heading in the document. Specifically, it:

- Adds an entry called `name` to the table of contents at the given level, with a hyperlink anchor at the current position.
- Updates the stored name for that level, as if a real heading had been created. For `chapter` to `subsection` this is done by calling the corresponding `mark` command, so any running headers are also updated.

The `level` must be one of `part`, `chapter`, `section`, `subsection` or `subsection`. Any other value results in an error.

```
\subsection{Real Subsection}
I believe I am in
  '\RpgGetName{subsection}'.

\RpgFake{subsection}{Fake Subsection}
Now I believe I am in
  '\RpgGetName{subsection}'
```

Real Subsection

I believe I am in 'Real Subsection'.
Now I believe I am in 'Fake Subsection'

Text Utilities

RPGTeX provides a number of commands for applying decorative styles to text and transforming it in ways useful for writing in an RPG context.

Any fonts or styles they apply are those assigned by the theme (page 30).

```
\RpgContour [<options>]{ text }
```

Typesets `text` with a coloured outline (contour), using the `contour` package. The following options are available:

Option	Effect	Default
<code>inner</code>	The colour of the text itself.	<code>ContourInnerColor</code> (page 10)
<code>outer</code>	The colour of the outline.	<code>ContourOuterColor</code>
<code>style</code>	Formatting commands (such as <code>\bfseries</code> or <code>\Huge</code>) applied to the text.	(empty)

Contours cannot automatically linebreak, instead the text is split into lines at each `\\`, and each word is contoured individually so that spacing is preserved. As a consequence, formatting commands placed within `text` may only affect the first word: all formatting should therefore be passed through the `style` key.

```
\RpgContour[inner=red,
  outer=black]{Contoured text}

\RpgContour[style=\Large\bfseries]{Styled
text \\ over two lines}
```

Contoured text
Styled text
over two lines

```
\RpgDice { expression }
```

Parses a dice expression of the form `<count>d<sides><modifier>` and typesets it in the format defined by the theme. The expression must satisfy the following rules:

- It contains a dice symbol (d or D), immediately followed by the number of sides.
- The dice symbol may be preceded by the number of dice. If it is omitted, the theme determines how the expression is displayed.
- Anything following the number of sides is treated as the modifier, which may be any numerical expression (such as `+3*2`). It is evaluated to a single number before being displayed.

Leading and trailing spaces are ignored. If the expression does not match this form, it is printed unchanged.

<code>\RpgDice{1d6-2}</code>	1d6-2
<code>\RpgDice{2D6 + 3*2^2}</code>	2d6+12
<code>\RpgDice{d8}</code>	d8
<code>\RpgDice{not a dice roll}</code>	not a dice roll

`\RpgDropCap` [`<options>`]{ `letter` }{ `text` }

Begins a paragraph with a large decorative initial (a ‘drop cap’), using the `lettrine` package. The `letter` is typeset in the `\RpgFontDropCap` (page 32) font element, and `text` -- intended to be the remainder of the first line -- in the `\RpgFontDropCapAfter` (page 32) font element.

The drop cap spans a number of lines set by the theme (four, by default), and is separated from the following text by the length `\RpgDropCapSeparation` (page 11). The optional argument accepts any `lettrine` options, which override these defaults.

<code>\RpgDropCap{T}{he example: this text runs over the first line, and then revert back to the normal font. It almost works! However, because it's wrapped in a text box, it goes slightly over the edges.}</code>	<i>T</i> he example: this text runs over the first line, and then revert back to the normal font. It almost works! However, because it's wrapped in a text box, it goes slightly over the edges.
--	--

`\RpgKey` { `text` }

`\emph` Typesets `text` in the `\RpgFontEmphasis` (page 31) font element, giving a consistent appearance to keywords (such as conditions or game terms). When the `font` option is active, and unless the `noemph` option is passed (page 8), `\emph` is redefined to be an alias of `\RpgKey`. This replaces its usual ‘context-aware’ behaviour (italics within upright text, and vice versa) with the keyword style.

<code>\emph{text}</code>	<i>text</i>
<code>\RpgKey{more text}</code>	<i>more text</i>

`\RpgOrdinal` [`<format>`]{ `number` }

Typesets an integer followed by its English ordinal suffix (`st`, `nd`, `rd` or `th`), correctly handling 11th, 12th and 13th. The optional `format` is a command applied to the suffix only, such as `\textbf`.

<code>\RpgOrdinal{1}, \RpgOrdinal{12}, \RpgOrdinal{23}</code>	1st, 12th, 23rd
<code>\RpgOrdinal[\textbf]{133}</code>	133rd

`\RpgPage` [`<style>`]{ `label` }

Typesets ``page'` followed by the page number of `label`, optionally enclosed in brackets. The `style` may be one of `t` (no brackets; the default), `p` (parentheses), `b` (square brackets) or `c` (curly braces). Any other value encloses the reference in question marks, to make the error easy to spot.

<code>\RpgPage{S:PackageOptions}</code>	page 8
<code>\RpgPage[p]{S:PackageOptions}</code>	(page 8)
<code>\RpgPage[b]{S:PackageOptions}</code>	[page 8]
<code>\RpgPage[c]{S:PackageOptions}</code>	{page 8}

`\RpgPlural` `(*)[<plural>]{ count }{ word }`

Typesets `count` followed by `word`, adding an ``s'` to the word unless `count` is 1. Irregular plurals may be given in the optional argument, which replaces the word whenever `count` is not 1. The starred form omits the count, printing only the word.

<code>\RpgPlural{1}{hat}, \RpgPlural{2}{hat}</code>	1 hat, 2 hats
<code>\RpgPlural[octopodes]{1}{octopus}, \RpgPlural[octopodes]{3}{octopus}</code>	1 octopus, 3 octopodes
<code>\RpgPlural*{3}{hat}</code>	hats

`\RpgSuperOrdinal` `{ number }`

Equivalent to `\RpgOrdinal[\textsuperscript]{ number }` (page 16), automatically applying a superscript style command.

<code>\RpgSuperOrdinal{42}</code>	42 nd
-----------------------------------	------------------

Full-Page Images

We use the `eso-pic` library for formatting whole-page images and placing content at specific positions on the page.

`\RpgPut` `[<width>][<format>](<anchor>){ x }{ y }{ content }`

Places `content` at a fixed position on the page, measured from the bottom-left corner. It may only be used within a page overlay, such as the first argument of `\RpgWholePageImage`.

The content is set in a paragraph of the given `width` (default: `\linewidth`), with `format` applied at the start (default: `\centering`).

The `anchor` determines which point of the content is placed at (x, y): `c` (centre; the default), `t`, `b`, `l`, `r`, or a combination such as `tl`.

`\RpgWholePageImage` `[<overlay>]{ path/to/image }[<fade>]`

Places an image so that it fills the whole of a new page, clearing the page on either side. This is useful for maps, handouts and illustrations.

The optional `overlay` is drawn on top of the image, and may contain any number of `\RpgPut` commands. The final optional argument (given in square brackets, after the path) fades the image towards white, as described for `\RpgCoverTransparency` (page 12).

Switches

Switches are named true/false values which can be used to change the content of a document without editing its text. For example, `RPGTeX` uses switches to decide whether secrets are shown, and whether rule environments are displayed as cards. All switches are global: changing a switch affects the whole of the remaining document.

Switches are designed to act as boolean counterparts to traditional `TeX` counters, and so have a similar interface.

The primary use-case for switches in `RPGTeX` is for the 'Cardable Environments', allowing the same document to typeset as both inline text, and as a 'playing card module' depending on the value of the associated switch.

```
\RpgIfSwitch { name }{ if-on }{ if-off }
```

Typesets `if-on` if the switch is on, and `if-off` otherwise.

```
\RpgNewSwitch { name }[<value>]
```

Creates a new switch called `name`, with an initial `value` (default: `false`). An error is raised if the switch already exists.

```
\RpgSwitch { name }{ value }
```

Sets an existing switch to the chosen value. Accepted input strings are:

Boolean State	Input Strings
True	on, On, true, True, 1
False	off, Off, false, False, 0

Any other value raises an error, as does a switch which does not exist.

```
\RpgToggle { name }
```

Turns an existing switch on if it is off, and off if it is on.

<pre>\RpgNewSwitch{HardMode} The dragon has \RpgIfSwitch{HardMode}{500}{200} HP. \RpgToggle{HardMode} The dragon has \RpgIfSwitch{HardMode}{500}{200} HP.</pre>	<pre>The dragon has 200 HP. The dragon has 500 HP.</pre>
--	--

Chapter 5: Environments

RPGTeX provides a number of environments for the elements which appear throughout RPG books: boxed text, tables, map keys, secrets and playing cards. Their appearance (fonts, colours and decoration) is set by the active theme, but may be adjusted through their options.

Many of these environments are built on the `tcolorbox` package. Unless stated otherwise, any option which is not recognised by an environment is passed on to `tcolorbox`, so all of its usual options (such as `colframe` or `breakable`) may also be used.

Callouts: RpgNarration, Sidebar & Tip

Callouts are boxed blocks of text which stand apart from the main text, the precise appearance determined by the theme. There are three types of callout, intended for different purposes:

RpgNarration	Text to be read aloud, such as the description of a scene at the start of an adventure.
RpgSidebar	Important information, such as key rules or summaries, which readers <i>should</i> pay attention to.
RpgTip	Helpful additions, such as tips and trivia, which are not essential but are too large for a footnote.

Each callout accepts the following options, in addition to the usual `tcolorbox` options:

Option	Effect	Default
color	The background colour of the box and its title, setting both <code>colback</code> and <code>colbacktitle</code> . An explicit <code>colback</code> or <code>colbacktitle</code> option takes priority over this value.	<code>NarrationColor</code> , <code>SidebarColor</code> or <code>TipColor</code> (page 10)
filigree	If present, decorates the box with the theme's filigree frame (page 21).	off

`\begin{ RpgNarration } [ <options> ]`

A box for text to be read aloud. It has no title, and its text is typeset in the `\RpgFontNarration` (page 32) font element on a `NarrationColor` background.

```
\begin{RpgNarration}[color=blue!30!white]
  This is text that you would read out
  loud to players, describing a scene.
  It will always be blue, even if the
  theme says otherwise -- because the
  optional argument takes priority.
\end{RpgNarration}
```

This is text that you would read out loud to players, describing a scene. It will always be blue, even if the theme says otherwise -- because the optional argument takes priority.

`\begin{ RpgSidebar } [ <options> ] { title }`

A box for important information, with a title. The title and body are typeset in the `\RpgFontSidebarTitle` (page 32) and `\RpgFontSidebarBody` (page 32) font elements, and the background is `SidebarColor`.

```
\begin{RpgSidebar}{A Sidebar}
  This is an important block of text,
  that you should pay attention to.
\end{RpgSidebar}
```

A Sidebar

This is an important block of text, that you should pay attention to.

`\begin{ RpgTip } [ <options> ] { title }`

A box for helpful additions, with a title. The title and body are typeset in the `\RpgFontTipTitle` (page 32) and `\RpgFontTipBody` (page 32) font elements, and the background is `TipColor`.

```
\begin{RpgTip}{A Tip}
  This is some helpful - but not vital -
  text.
\end{RpgTip}
```

A Tip

This is some helpful - but not vital - text.

RpgTables

`\RpgRowStyle { format }`

Applies **format** to every cell of the current row of an **RpgTable**. It must be placed at the start of the row. This is the mechanism used to style the header row, and may be used to highlight any other row in the same way.

`\begin{ RpgTable } [<options>]{ columns }`

Typesets a table in the style of the theme. By default this is a `tabularx` table which fills the width of the line, so the usual column types (`l`, `c`, `r`, `p{<width>}`, ...) may be used alongside the stretching `X` column. **RPGTeX** also provides a centred stretching column, `Y`.

By default, the table:

- Typesets its first row as a header, in the `\RpgFontTableHeader` (page 32) font element, and the remaining rows in `\RpgFontTableBody` (page 32).
- Colours alternate rows in `TableColor` (page 10), beginning with the second row.

The following options are available:

Option	Effect	Default
<code>breakable</code>	If present, uses <code>xltabular</code> , allowing the table to break across pages. This is only possible in one-column mode.	off
<code>color</code>	The colour used for the coloured rows.	<code>TableColor</code> (page 10)
<code>color-both</code>	If present, colours every row.	off
<code>color-odd</code>	If present, colours the odd rows (including the header) instead of the even rows.	off
<code>filigree</code>	If present, wraps the table in an <code>RpgFiligree</code> (page 21).	off
<code>filigree-hgap</code>	The horizontal padding between the filigree and the table.	1pt
<code>filigree-opacity</code>	The opacity of the filigree's background.	1
<code>headskip</code>	The vertical space below the title.	5pt plus 2pt minus 2pt
<code>intro</code>	Introductory text, typeset between the title and the table.	(empty)
<code>introskip</code>	The vertical space below the introduction.	5pt plus 2pt minus 2pt
<code>noheader</code>	If present, the first row is typeset in the same style as the rest of the table.	off
<code>simple</code>	If present, uses a plain <code>tabular</code> , whose width is determined by its contents. The <code>width</code> , <code>breakable</code> and <code>noheader</code> options are then ignored, and the first row is not styled as a header.	off
<code>title</code>	A title, typeset above the table in the <code>\RpgFontTableTitle</code> (page 32) font element.	(empty)
<code>vskip</code>	The vertical space above and below the table.	9pt plus 3pt minus 3pt
<code>width</code>	The width of the table.	<code>\linewidth</code>

```

\begin{RpgTable}[width=0.85\linewidth,
color=green!30!white]{lX}
  Header 1 & Header 2
  \\
  Text & Text fills up to 85\% of the
    line width then breaks
  \\
  Alternating & This row is transparent
  \\
  Colour & but this one is green
\end{RpgTable}

%suppress header formatting, add title and
  filigree:
\begin{RpgTable}[title={Test
Table},noheader, filigree]{XlX}
  Plain & Header & Text
  \\
  Now there's no & difference & between the
    the header and the main
  \\
  body
\end{RpgTable}

```

Header 1	Header 2
Text	Text fills up to 85% of the line width then breaks
Alternating	This row is transparent
Colour	but this one is green

Test Table

Plain	Header	Text	
Now there's no	difference	between the	header and
body		the main	

RpgAbstracts

```
\begin{ RpgAbstract } [<title>] [<width>] [<gap>]
```

abstract Typesets a short, centred summary block, such as the introduction to an adventure. The **title** (default: ``Abstract'`) is typeset in the `\RpgFontAbstractTitle` (page 32) font element, and the body in `\RpgFontAbstractBody` (page 32). The block has the given **width** (default: `0.7\linewidth`), with a vertical **gap** above and below it (default: 15pt). In two-column mode, the block spans both columns. The standard **abstract** environment is redefined to be identical to **RpgAbstract**.

```
\begin{ summary } [<title>] [<width>] [<gap>]
```

Identical to **RpgAbstract**, but with the default title ``Summary'`.

RpgFiligree

Filigree is a decorative frame, drawn around the outside of a box in place of its usual border. Its design is provided by the theme, and it is drawn in **FiligreeColor** (page 10).

```
\begin{ RpgFiligree } [<options>]
```

A plain box decorated with the filigree frame, with a **FiligreeBackColor** background. It is equivalent to an untitled **RpgTip** (page 19) with the **filigree** option, and accepts the same options.

The filigree may also be added to any **tcolorbox** by including the **rpgfiligree** style in its options, or to a callout or table with the **filigree** option.

```

%% As environment
\begin{RpgFiligree}
  A basic filigree-wrapped tbox.
\end{RpgFiligree}

%% As option
\begin{RpgSidebar}[filigree]{Fancy Sidebar}
  A very fancy sidebar
\end{RpgSidebar}

```

A basic filigree-wrapped tbox.

Fancy Sidebar
A very fancy sidebar

RpgSecrets

Game Masters often need notes which their players should not see. The `RpgSecret` environment allows the same document to serve both purposes, by marking text as secret and choosing whether to reveal it.

```
\begin{ RpgSecret } [<options>]{<public-text>}
```

Marks the body of the environment as secret. Its appearance depends on the `ShowSecrets` switch (page 18), which is off by default:

Switch	Display
Off	Only the (optional) <code>public-text</code> is shown, as ordinary text, with nothing to suggest that anything has been withheld.
On	The secret text is shown in a highlighted box, labelled as hidden. If <code>public-text</code> was given, it is shown in the lower part of the box, labelled as the publicly available version.

Secrets are revealed for the remainder of the document with `\RpgSwitch{ ShowSecrets }{ on }`. The following options are available:

Option	Effect	Default
<code>color</code>	The colour of the box's frame.	<code>SecretColor</code>
<code>label-public</code>	The label above the public text.	<code>`Normal text'</code>
<code>label-secret</code>	The label above the secret text.	<code>`Hidden text'</code>
<code>switch</code>	The switch which controls this secret. Using a different switch allows separate groups of secrets to be revealed independently.	<code>ShowSecrets</code>

The labels are typeset in the `\RpgFontSecretTitle` (page 32) font element.

RpgSecret: Hidden

```

\RpgSwitch{ShowSecrets}{false}

\lorumIpsumOne%Snippets of standard lorem ipsum
\begin{RpgSecret}{}
  The players don't know it yet, but all this
  pseudo-latin text is a hint that devils are
  involved!
\end{RpgSecret}
\lorumIpsumTwo{}
\begin{RpgSecret}{
  Logos kykloforia, thelxis kai ethos, tyrannis
  kai epilysis. Aei theria plaka, kyrios meion
  egestas, anagkas proin eira kai theria.
}

  This bit of text is noteworthy - it's been
  added in after the fact
\end{RpgSecret}
\lorumIpsumThree

```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Logos kykloforia, thelxis kai ethos, tyrannis kai epilysis. Aei theria plaka, kyrios meion egestas, anagkas proin eira kai theria. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc.

```
%exactly the same as above, but reveal the secrets!
\RpgSwitch{ShowSecrets}{true}

\lorumIpsumOne%Snippets of standard lorem ipsum
\begin{RpgSecret}{
  The players don't know it yet, but all this
  pseudo-latin text is a hint that devils are
  involved!
}
\end{RpgSecret}
\lorumIpsumTwo{
\begin{RpgSecret}{
  Logos kykloforia, thelxis kai ethos, tyrannis
  kai epilysis. Aei theria plaka, kyrios meion
  egestas, anagkas proin eira kai theria.
}

  This bit of text is noteworthy - it's been
  added in after the fact
}
\end{RpgSecret}
\lorumIpsumThree
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.

Hidden text

The players don't know it yet, but all this pseudo-latin text is a hint that devils are involved!

Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Hidden text

This bit of text is noteworthy - it's been added in after the fact

Normal text

Logos kykloforia, thelxis kai ethos, tyrannis kai epilysis. Aei theria plaka, kyrios meion egestas, anagkas proin eira kai theria.

Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc.

RpgMaps

Adventures often describe a location as a numbered list of areas, which correspond to the labels on a map. The `RpgMap` environment produces such a list, numbering and nesting the areas automatically.

```
\begin{ RpgMap } [<options>]
```

Begins a list of map areas, each of which is created with `\area`. Each area is given a heading, whose size depends on how deeply the map is nested: `\section*` at the top level, then `\subsection*`, and so on, down to `\subparagraph*`. These headings do not appear in the table of contents.

Areas are numbered according to their depth, cycling through numbers, capital letters and lower-case Roman numerals, so that the second area within the first area is 1B, and its third sub-area is 1B-iii. The following options are available:

Option	Effect	Default
<code>empty-prefix</code>	Text placed before the number of an unnamed area.	<code>`Area '</code>
<code>header-offset</code>	The number of levels by which to shrink the headings.	0
<code>prefix</code>	Text placed before the number of a named area.	<code>`Area '</code>
<code>title</code>	A title for the map, typeset as a heading one level larger than its areas. It is only used by the outermost map.	(empty)

```
\RpgMapRef { label }
```

`\RpgMapRef*` Typesets the name of the labelled area. The starred form typesets its number instead. This is a wrapper around `\ref` and `\nameref`.

```
\RpgMapRef{map:dais} is area
\RpgMapRef*{map:dais}, on
\RpgPage{map:dais}.
```

The Dais is area 1A, on page 24.

Inside a Map

Inside a map environment, the following commands and environments are valid.

`\area [<label>]{<name>}`

Adds an area to the current map, much as `\item` adds an item to a list. Its heading is ``<prefix><number>: <name>'`, or ``<empty-prefix><number>'` if no `name` is given. Any text which follows forms the description of the area.

If a `label` is given, the area may be referred to with `\RpgMapRef` or `\RpgPage`.

`\begin{RpgArea} [<label>]{<name>}`

Adds an area to the current map, and begins a nested map within it, so that the body of the environment may contain sub-areas. It is equivalent to an `\area` followed by an `RpgMap` environment, and raises an error if used outside of a map.

Nested Map Areas

```
\begin{RpgMap}[header-offset=2]
  \begin{RpgArea}[map:hall]{Great Hall}
    A long and draughty hall.

    \area[map:dais]{The Dais} A raised
      stone platform.

    \area An unremarkable alcove.
  \end{RpgArea}

  \area[map:cellar]{Cellar} Damp and
    dark.
\end{RpgMap}
```

Area 1: Great Hall

A long and draughty hall.

Area 1A: The Dais. A raised stone platform.

Area 1B. An unremarkable alcove.

Area 2: Cellar

Damp and dark.

RpgCards

Cards are small, playing-card sized boxes, useful for handouts such as items, spells or abilities. If the content of a card is too long to fit, it is automatically continued onto further cards.

Cards are rarely used directly: they are most often created by the rule environments (page 26), which can display the same content either as normal text or as a card.

RpgCard example

```
\begin{RpgCard}[vmargin=0.1cm, hmargin=0.1cm,
  height=5.2cm, width=3.6cm]
  \raggedright\footnotesize This is an example
  card with lots of text\footnote{And maybe a
    footnote too?}.
  \lipsum[2]
\end{RpgCard}
```

This is an example card with lots of text*. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis.

Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

*And maybe a footnote too?

`\cardbreak {}`

Within an `RpgCard`, ends the current card and continues the text on the next, in the same way as `\pagebreak`. Outside of a card it has no effect, so it may safely be used in text which is only sometimes displayed as a card.

`\footnote { text }`

Within an `RpgCard`, `\footnote` is redefined so that its text appears at the end of the card's contents, below a horizontal rule. **Only one footnote may appear in each card:** if more than one is used, only the last is shown. Card footnotes do not affect the document's footnote numbering.

`\begin{ RpgCard } [ <options> ]`

Typesets its contents in one or more cards, placed side by side. The text is typeset in the `\RpgFontCardBody` (page 32) font element. The following options are available:

Option	Effect	Default
<code>cardsep</code>	The horizontal space between consecutive cards.	5pt
<code>color</code>	The colour of the card's background.	white
<code>footnote</code>	The symbol used to mark the card's footnote.	*
<code>height</code>	The outer height of each card.	9.7cm
<code>hmargin</code>	The horizontal space between the edge of the card and its text. It must be less than half of the width.	0.2cm
<code>opacity</code>	The opacity of the card's background.	1
<code>vcenter</code>	An alias of <code>vcentre</code> .	off
<code>vcentre</code>	If present, centres the text vertically within each card.	off
<code>vmargin</code>	The vertical space between the edge of the card and its text. It must be less than half of the height.	0.2cm
<code>width</code>	The outer width of each card.	6.75cm

Cards cannot be given a title, and cannot be nested within one another.

```
\begin{tikzpicture}%wrap in tikz so we can
draw dimensions
\node[anchor=south west,inner
sep=0pt,outer sep=0pt] at (0,0)
{
  \begin{RpgCard}[hmargin=0.5cm,
vmargin=0.5cm, height=9.7cm,
width=6.75cm]
    \subsubsection{Text goes here}
    It fits nicely into the card,
    wrapping over
    lines\footnote{We can also
    have a single footnote}.
  \end{RpgCard}\ignorespaces
};
%% Draw the dimenions on top; use macros
to save space!
\heightLines{red}
\widthLines{ForestGreen}

\end{tikzpicture}
```

`\RpgCardReset {}`

Restores the default options of all subsequent cards to their original values.

`\RpgCardSet { options }`

Sets the default options of all subsequent cards. Options given to an individual `RpgCard` take priority over these defaults.

Rule Environments

The environments described so far are independent of any particular game system. Some elements, however, depend on the rules of the game being written for: a monster's statistics in one system look nothing like those in another. **RPGTeX** therefore provides four *rule environments*, whose properties and appearance are defined by the theme:

Environment	Purpose
RpgFeat	Feats, talents and other character features.
RpgItem	Items and equipment.
RpgSpell	Spells and similar abilities.
RpgStat	Statistics blocks for creatures and characters.

Each may be displayed either as normal text or as a card, controlled by a switch. Since the available properties depend on the theme, consult the documentation of the active theme for the details of each.

`\RpgGetKey` (Variable)

The value given to a property of the current rule environment, such as `\RpgGetKey{ rarity }`. It may only be used within a rule environment.

```
\begin{RpgItem}{<name>}[<key> = <value>]
  The value of the key is \RpgGetKey{<key>}.
\end{RpgItem}
```

```
\begin{ RpgIfEnv } { switch }[<options-on>]{ env-on }[<options-off>]{ env-off }
```

Begins the environment **env-on** (with **options-on**) if **switch** is on, and **env-off** (with **options-off**) otherwise. Any arguments which follow are passed to the chosen environment, so both environments must take the same arguments. This is the mechanism by which the rule environments switch between text and card modes, and it may be used to create similar switchable environments.

```
\begin{ Rpg[X] } [<card-options>]{ name }[<properties>]
```

Typesets the rule element 'X' called **name**, with the given **properties** (a list of key-value pairs defined by the theme). Its display depends on the corresponding switch (page 18):

Environment (X)	Switch
RpgFeat	UseFeatCard
RpgItem	UseItemCard
RpgSpell	UseSpellCard
RpgStat	UseStatCard

If the switch is on, the element is displayed as an **RpgCard** (page 25), to which the **card-options** are passed. Otherwise (the default), it is displayed as normal text, and the **card-options** are ignored.

PART II

RPGTeX For Designers

Chapter 6: Building a Theme

This part of the manual documents the commands used to build themes. Authors of **RPGTeX** documents will rarely need them; they are intended for designers creating or modifying a theme, and for users who wish to adjust an existing theme in detail. By convention, all of these commands begin with **RpgTheme**.

This chapter describes how a theme is organised. The chapters which follow describe how a theme defines its fonts (page 30), the design of its pages (page 33), the appearance of individual elements (page 35), and its rule environments (page 37).

Code Setters

Many of the commands in this part of the manual do not perform any action themselves. Instead, they *store* a piece of code, which **RPGTeX** runs later, when the corresponding element is created. For example, **\RpgThemePartPage** stores the code used to draw a part page, which is run each time **\part** is called.

The stored code is often given arguments at the time it is run, such as the name of the part. These are accessed as **#1**, **#2** and so on within the stored code, even though the setter itself is called with only a single argument:

```
\RpgThemePartPage{The part is called #1}
```

The entry for each setter lists the arguments which are available.

Theme Files

A theme called **name** is a directory, **<themepath>/name**, containing the file **name.cfg**. Calling **\RpgSetTheme** (page 12){**name** } inputs this file, which in turn usually inputs a number of **.theme.tex** files, each customising one part of the document (colours, fonts, headers, titles and so on). For example, the default theme's configuration file begins:

```
\def\base{themes/default}  
\input{\base/colors.theme.tex}  
\input{\base/callouts.theme.tex}  
...
```

Since a theme may be loaded in the middle of a document, its commands should only change appearance, and each theme should set every value it relies upon, rather than assuming the defaults of the theme before it.

Conditional Code

Many parts of a theme are only meaningful when the corresponding package option is active. The following commands allow a theme to include code conditionally:

```
\RpgIfFont { code }
```

Runs code only if the **font** option is active (page 8).

```
\RpgIfPaper { code }
```

Runs code only if the **background** option is active.

```
\RpgIfSections { code }
```

Runs code only if the **sections** option is active.

```
\RpgIfTitle { code }
```

Runs `code` only if the `title` option is active.

```
\RpgThemeLayoutOnly { code }
```

Runs `code` only if `RPGTeX` was loaded through one of its classes, rather than directly as a package. This is used for changes to the overall layout of the document (such as hyphenation and list spacing), which should not be imposed on a document of another class.

Chapter 7: Fonts

RPGTeX allows a high degree of customisation of the fonts and typefaces used throughout a document. Rather than setting fonts directly, each part of the document (headings, table text, footers and so on) draws its font from a named *font element*, such as `\RpgFontSection` (page 32). The theme defines the appearance of each element using `\RpgThemeFont`, and assigns each element to the relevant part of the document. Changing the appearance of an element therefore changes it consistently wherever that element is used.

Why didn't my font change?

`\RpgThemeFont` does not change any fonts directly: it updates the stored definition of a *font element*, which a theme designer must then assign to the relevant part of the document. For example, the `\RpgFontSection` (page 32) element does not 'hook in' to the `\section` command by itself; it changes the section font only because most themes call `\titleformat{\section}{\RpgFontSection}...`. If the `\titleformat` were later changed (for whatever reason), the section font would no longer update.

If an element does not change font after its definition is updated, first check that the font element is actually being invoked -- and if not, invoke it manually. Once the command is in place, the font will change as expected.

Note also that `\RpgThemeFont` has no effect unless the `font` option is active (page 8). This is the default when using one of the RPGTeX classes.

Font Commands

```
\RpgThemeFont { key-value-list }
```

Updates the font elements specified by a comma-separated list of key-value pairs. Each element has two keys, `<name>-family` and `<name>-style`, where `<name>` is given in the 'Components' column of the table below (or was assigned by `\RpgThemeCreateFont`).

The values may be any L^AT_EX font or formatting commands (such as `\slshape\scriptsize\bfseries`), which are applied whenever the element is invoked. To change the typeface itself, a font family must first be defined, for example using `\newfontfamily` from the `fontspec` package.

Font Interface

```
\subsection{The Original Font}
Here is some text
```

```
\newfontfamily{\courier}{Courier}
\RpgThemeFont{main-body-family=\courier,
subsection-style=\slshape\Huge}
```

```
\subsection{The New Font}
And after the change is introduced
```

The Original Font

Here is some text

The New Font

And after the change is introduced

```
\RpgThemeCreateFont { \ElementName }{ key-base }
```

Creates a new font element, `\ElementName`, along with the keys `<key-base>-family` and `<key-base>-style`. A new element is initially blank (it applies no formatting) until it is assigned a value using `\RpgThemeFont`. Thereafter, it may be used and modified in exactly the same way as the built-in font elements.

Create Font

```
\RpgThemeCreateFont{\MyFont}{myfont}
\newfontfamily{\courier}{Courier}
\RpgThemeFont{
  myfont-family = \courier,
  myfont-style = \slshape
}

This is text

{\MyFont This is text in my new font}
```

This is text
This is text in my new font

Font Elements

RPGTeX provides 35 font elements by default (themes may define more). Since each element is assigned to part of the document by the theme designer, an element intended for one purpose (such as section headings) may be used for a different purpose within another theme. The descriptions below therefore reflect how the elements are used by RPGTeX itself and by the provided themes.

Family vs Style

Each font element is defined by two components: a **family** and a **style**. Formally, the **family** defines the **typeface** used by the element, whilst the **style** determines the options applied to that typeface (weight, shape, size, colour and so on).

In practice the distinction is largely one of clarity, since the font element simply applies the two in sequence:

```
\RpgFontX = <x-family> <x-style>
```

It is therefore generally safe to place commands which belong in the family within the style key, provided that they do not conflict with other styling.

Font vs Implementation

We encourage designers to place all text styling within the relevant font element, rather than elsewhere. If all subsections are to be red, then set `subsection-style=\color{ red }`, rather than setting the colour within the `titlesec` specification.

There will naturally be some exceptions to this: we found that the `RpgTitleFont` colour we wanted within `RpgDrawCover` diverged so strongly from that in `RpgSimpleTitle` that it made sense to define a special colour when rendering over a background image.

Font Element	Components	Usage
\RpgFontBody	main-body-family main-body-style	The main body text of the document. \RpgThemeFont sets \normalfont equal to this element. Changing the font size here (e.g. with \large) can give counterintuitive results, since it affects <i>only</i> the body text and does not scale the other elements accordingly. The font size of the whole document should instead be set using the <code>size</code> option (page 8).
\RpgFontEmphasis	emph-family emph-style	The font used by \RpgKey, and therefore by \emph unless the <code>noemph</code> option is passed.
\RpgFontTitle	title-family title-style	The font used for the document title, as returned by \RpgGetTitle and displayed by \maketitle.
\RpgFontSubTitle	subtitle-family subtitle-style	The font used for the subtitle and author, as returned by \RpgGetSubtitle and \RpgGetAuthor and displayed by \maketitle.
\RpgFontPart	part-family part-style	The font used for the part name on the part page created by \part.
\RpgFontChapter	chapter-family chapter-style	The font used for \chapter headings.

<code>\RpgFontSection</code>	section-family section-style	The font used for <code>\section</code> headings.
<code>\RpgFontSubsection</code>	subsection-family subsection-style	The font used for <code>\subsection</code> headings.
<code>\RpgFontSubsubsection</code>	subsubsection-family subsubsection-style	The font used for <code>\subsubsection</code> headings.
<code>\RpgFontParagraph</code>	paragraph-family paragraph-style	The font used for <code>\paragraph</code> headings.
<code>\RpgFontSubparagraph</code>	subparagraph-family subparagraph-style	The font used for <code>\subparagraph</code> headings.
<code>\RpgFontTableTitle</code>	table-title-family table-title-style	The title of an <code>RpgTable</code> (page 20), if one is set using the <code>title</code> option.
<code>\RpgFontTableHeader</code>	table-header-family table-header-style	The first (header) row of an <code>RpgTable</code> (page 20).
<code>\RpgFontTableBody</code>	table-body-family table-body-style	The rows of an <code>RpgTable</code> (page 20) after the header row.
<code>\RpgFontTipTitle</code>	tip-title-family tip-title-style	The title of an <code>RpgTip</code> (page 19) environment.
<code>\RpgFontTipBody</code>	tip-body-family tip-body-style	The body of an <code>RpgTip</code> (page 19) environment.
<code>\RpgFontSidebarTitle</code>	sidebar-title-family sidebar-title-style	The title of an <code>RpgSidebar</code> (page 19) environment.
<code>\RpgFontSidebarBody</code>	sidebar-body-family sidebar-body-style	The body of an <code>RpgSidebar</code> (page 19) environment.
<code>\RpgFontNarration</code>	narration-family narration-style	All text within an <code>RpgNarration</code> (page 19) environment (which has no title).
<code>\RpgFontAbstractTitle</code>	abstract-title-family abstract-title-style	The title of an <code>RpgAbstract</code> (page 21) environment.
<code>\RpgFontAbstractBody</code>	abstract-body-family abstract-body-style	The body of an <code>RpgAbstract</code> (page 21) environment.
<code>\RpgFontSecretTitle</code>	secret-title-family secret-title-style	The labels which distinguish the hidden and public text of an <code>RpgSecret</code> (page 22) environment when secrets are shown.
<code>\RpgFontCardTitle</code>	card-title-family card-title-style	The title of a card, as used by the card-switching environments of the provided themes (such as <code>RpgItem</code>).
<code>\RpgFontCardHeader</code>	card-header-family card-header-style	Subheadings and descriptive headers within a card (such as the type and rarity of an <code>RpgItem</code>).
<code>\RpgFontCardBody</code>	card-body-family card-body-style	The body text of an <code>RpgCard</code> (page 25).
<code>\RpgFontTocPart</code>	toc-part-family toc-part-style	Part entries in the table of contents.
<code>\RpgFontTocChapter</code>	toc-chapter-family toc-chapter-style	Chapter entries in the table of contents.
<code>\RpgFontTocSection</code>	toc-section-family toc-section-style	Section entries in the table of contents.
<code>\RpgFontStatBlockTitle</code>	stat-block-title-family stat-block-title-style	The title of a <code>statblock</code> environment (in the dnd theme, <code>RpgStat</code>).
<code>\RpgFontStatBlockBody</code>	stat-block-body-family stat-block-body-style	The body text of a <code>statblock</code> environment.
<code>\RpgFontStatBlockSection</code>	stat-block-section-family stat-block-section-style	Section headings within a <code>statblock</code> environment.
<code>\RpgFontFooter</code>	footer-family footer-style	The footer text.
<code>\RpgFontPageNumber</code>	page-number-family page-number-style	The page number within the footer.
<code>\RpgFontDropCap</code>	drop-cap-family drop-cap-style	The large initial letter created by <code>\RpgDropCap</code> .
<code>\RpgFontDropCapAfter</code>	drop-cap-after-family drop-cap-after-style	The text which follows the initial letter of an <code>\RpgDropCap</code> , on the first line.

Chapter 8: Page Design

This chapter describes the commands which define the overall design of a theme's pages: their backgrounds and footers, and the title and part pages.

Backgrounds & Footers

These commands control the page background and footer. Their effects depend on the **bg** option (page 8): the paper colour and image are only shown when it is **full**, and the footer decorations unless it is **none**. The paper and footer commands have no effect unless the **background** option is active.

The footer commands draw on the page using the **eso-pic** package. Positions are measured from the bottom-left corner of the page, and further content may be placed with `\RpgPut` (page 17).

`\RpgThemeFooterDecoration` **(*)**{ height }{ path/to/image }{ overlay }

Draws an image across the full width of the bottom of the current page, with the given **height**, and then draws the **overlay** (such as `\RpgPut` or `\RpgThemePageNumberAt` commands) on top of it.

The decoration is only drawn on the page on which the command is called, and so it is normally placed within the definition of a page footer. The starred form mirrors the image horizontally, and the unstarred form draws it as it is, allowing a single image to be used on both left- and right-hand pages:

```
\fancyfoot[LE]{%
  \RpgThemeFooterDecoration{1.5cm}{path/to/scroll}{
    \RpgPut{0.43in}{0.15in}{\RpgFontPageNumber \thepage}
  }
}
\fancyfoot[RO]{%
  \RpgThemeFooterDecoration*{1.5cm}{path/to/scroll}{
    \RpgPut{\paperwidth-0.43in}{0.15in}{\RpgFontPageNumber \thepage}
  }
}
```

`\RpgThemePageNumberAt` { x }{ y }

Typesets the page number, in the `\RpgFontPageNumber` (page 32) font element, centred at the position (x, y). It may only be used within a page overlay, such as the final argument of `\RpgThemeFooterDecoration`.

`\RpgThemePaperColor` { colour }

Sets the background colour of every page, and updates `PaperColor` to match (page 10).

`\RpgThemePaperImage` { path/to/image }

Sets an image, stretched to fill the page, as the background of every subsequent page. This is typically a paper texture. An empty path removes any existing background image.

Title & Part Pages

These commands define the appearance of the pages created by `\maketitle` (page 13) and `\part` (page 13). The cover and part page designs are drawn on the page with **eso-pic**, usually with `\RpgPut` (page 17), while the header title is typeset as ordinary text. Within them, the title information is available through `\RpgGetTitle`, `\RpgGetSubtitle` and `\RpgGetAuthor` (page 12).

`\RpgThemeOnPartChange { code }`

Stores `code` to be run after every `\part` (in either form). For example, the `dnd` theme uses this to cycle its colour scheme with each new part.

`\RpgThemePartPage { code }`

Stores the design of the part page created by `\part`, which is drawn on top of the part's background image (if any). The available arguments are:

Argument	Value
----------	-------

#1	The name of the part.
----	-----------------------

The part number is available through `\thepart`. By default, the part number and name are centred on the page in the `\RpgFontPart` (page 31) font element.

`\RpgThemeTitleCover { code }`

Stores the design of the cover page created by `\maketitle` in `cover` mode. It is drawn on top of the `\cover` image, if one has been set, or on an otherwise blank page.

`\RpgThemeTitleHeader { code }`

Stores the code which typesets the title when `\maketitle` is called in `header` mode. Unlike the cover, this is ordinary text, placed at the point where `\maketitle` is called.

`\RpgThemeTitleMode { cover / header }`

Selects which of the two designs `\maketitle` produces. Any other value raises an error. The `rpgbook` class selects `cover`; otherwise, the default is `header`.

Chapter 9: Styling Elements

The appearance of most environments is controlled through `tcolorbox` styles, which a theme may redefine with `\tcbset`:

```
\tcbset{rpgsidebar/.style={colframe=black, ...}}
```

The available styles are `rpgnarration`, `rpgsidebar` and `rpgtip` (the callouts), `rpgsecret` (the box used by `RpgSecret` (page 22)) and `rpgcard` (the cards). The following commands control the elements which are not built from a `tcolorbox` style.

`\RpgThemeDiceFormat { code }`

Stores the code which typesets a dice expression, after it has been parsed by `\RpgDice` (page 15). The available arguments are:

Argument	Value
#1	The number of dice (empty if none was given).
#2	The number of sides.
#3	The evaluated modifier, including its sign (for example, +6 or -2).

By default, the expression is typeset as `#1d#2#3`.

The arguments are passed as variables which contain these values, rather than as the values themselves. They typeset correctly as they are, but must be expanded before being tested or compared: for example, with `\tl_if_blank:eTF` rather than `\tl_if_blank:nTF`.

RpgDice Formatting: The D&D Format

Before: `\RpgDice{2d8 + 3}`, `\RpgDice{d8}`

```
\ExplSyntaxOn
\RpgThemeDiceFormat{ % #1: Dice Number,
  #2: Dice Sides, #3: Modifier
  %% d8 -> 1d8
  %% the arguments are variables, so
  %% must be expanded before testing
  \tl_set:Nx \l_tmpa_tl {
    \tl_if_blank:eTF {#1} {1} {#1} }
  %% compute the average roll, and print
  %% it first
  \fp_eval:n { floor ( \l_tmpa_tl * ( #2
    + 1 ) / 2 ) + ( #3 ) }
  ~( \l_tmpa_tl d#2 \fp_compare:nNnF {
    #3 } = { 0 } { #3 } )
}
\ExplSyntaxOff
```

After: `\RpgDice{2d8 + 3}`, `\RpgDice{d8}`

Before: 2d8+3, d8
After: 12(2d8+3), 4(1d8)

`\RpgThemeDropCapHeight { lines }`

Sets the number of lines spanned by the initial letter of `\RpgDropCap` (page 16). The default is 4.

`\RpgThemeMapFormat { code }`

Stores the code which formats the number of an area within an `RpgMap` (page 23). The available arguments are:

Argument	Value
----------	-------

#1	The depth of the area (1 for the outermost map).
#2	The position of the area at that depth (1 for the first area).

The full number of an area is formed by joining the formatted numbers at each depth. By default, the depths cycle through numbers, capital letters, and lower-case Roman numerals preceded by a hyphen, giving numbers such as 1B-iii.

`\RpgThemeSetFiligree { code }`

Stores the `tikz` code which draws the filigree frame (page 21). It is drawn over the box, with `FiligreeColor` as the line colour, and may refer to the box's corners through the `frame` node (such as `frame.north west`). The size of the design should be scaled by `\RpgFiligreeSize` (page 11).

`\RpgThemeSetSecretHidden { code }`

Stores the code which typesets an `RpgSecret` while secrets are hidden. The available arguments are:

Argument	Value
----------	-------

#1	The public text (possibly empty).
#2	The secret text.
#3	Any options given to the environment which are not recognised by it, intended to be passed to a <code>tcolorbox</code> .

By default, only the public text (**#1**) is typeset.

`\RpgThemeSetSecretVisible { code }`

Stores the code which typesets an `RpgSecret` while secrets are shown. The available arguments are the same as for `\RpgThemeSetSecretHidden`. By default, the secret text is typeset in a `tcolorbox` with the `rpgsecret` style, with the public text (if any) in its lower part.

Chapter 10: Rule Environments

Rule environments (page 26) are the elements of a document which depend on the rules of a particular game, such as items, spells and monsters. `RPGTeX` itself only creates them: the theme decides which properties each one accepts, and how it is displayed. `RPGTeX` creates four rule environments, `RpgItem`, `RpgSpell`, `RpgFeat` and `RpgStat`, each of which may be displayed either as text or as a card. Designers may also create their own.

Defining a rule environment has three steps:

1. **Create** the environment, with `\RpgThemeNewRuleEnv` or `\RpgThemeNewCardableRuleEnv`. This is only necessary for new environments, since the four standard ones already exist.
2. **Add properties** to it, with the `\Rpg[X]AddProperty` and `\Rpg[X]AddBoolean` commands generated for each environment.
3. **Set its format**, with the `\RpgTheme[X]Format` (and, for a card, `\RpgTheme[X]CardFormat`) command generated for each environment.

Throughout this chapter, `[X]` stands for the name of the environment, so that the commands for `RpgItem` are `\RpgItemAddProperty`, `\RpgThemeItemFormat` and so on.

Creating a Rule Environment

```
\RpgThemeNewCardableRuleEnv{Potion}{UsePotionCard}
\RpgPotionAddProperty{colour}{\PotionColour}{clear}

\RpgThemePotionFormat{
  \paragraph{#1} (\PotionColour) #2
}
\RpgThemePotionCardFormat{
  {\RpgFontCardTitle #1}\par
  \textit{A \PotionColour{} liquid}\par
  #2
}

\begin{RpgPotion}{Healing Draught}[colour=red]
  Restores \RpgDice{2d4+2} hit points.
\end{RpgPotion}

\RpgSwitch{UsePotionCard}{on}
\begin{RpgPotion}[width=4.5cm,
  height=3cm]{Healing Draught}[colour=red]
  Restores \RpgDice{2d4+2} hit points.
\end{RpgPotion}
```

Healing Draught. (red) Restores 2d4+2 hit points.

Healing Draught

A red liquid

Restores 2d4+2 hit points.

Creating Rule Environments

`\RpgThemeNewCardableRuleEnv` [`<prefix>`]{ `name` }{ `switch` }[`<card-options>`][`<properties>`]

Creates the environment `<prefix><name>` (the prefix is `Rpg` by default), which may be displayed either as text or as a card, together with a new switch (page 18) which chooses between them. The switch is off (text mode) by default. The environment is used as:

```
\begin{<prefix><name>}[<card-options>]{<title>}[<properties>]
  <body>
\end{<prefix><name>}
```

In card mode, the `card-options` are passed to `RpgCard` (page 25); in text mode they are ignored. The optional `card-options` and `properties` given to `\RpgThemeNewCardableRuleEnv` are the defaults used when the corresponding argument is omitted. They are replaced, rather than combined, when the argument is given.

In addition to the commands generated by `\RpgThemeNewRuleEnv`, this creates `\RpgTheme[X]CardFormat`.

`\RpgThemeNewRuleEnv` [`<prefix>`]{ `name` }

Creates the environment `<prefix><name>` (the prefix is `Rpg` by default), which is always displayed as text. The environment is used as:

```
\begin{<prefix><name>}{<title>}[<properties>]
  <body>
\end{<prefix><name>}
```

It also creates the commands `\Rpg[X]AddProperty`, `\Rpg[X]AddBoolean` and `\RpgTheme[X]Format`, using the same prefix.

Properties

Each property is a key which may be given in the `properties` argument of the environment. Its value is stored in a variable, which the format can then use. Properties are set separately for each use of the environment, so a property which is not given takes its default value.

Properties belong to the environment, not to the theme which added them. Adding a property with the same name again replaces it, but a property added by one theme remains available after another theme is loaded.

`\Rpg[X]AddBoolean` (*){ `key` }{ `variable` }

Adds a flag property, which takes no value. The unstarred form sets the boolean `variable` to false, and makes it true when the flag is given; the starred form does the opposite.

`\Rpg[X]AddProperty` { `key` }[`<aliases>`]{ `variable` }{ `default` }

Adds a property which takes a value, stored in `variable`, with the given `default`. The optional `aliases` are a comma-separated list of alternative names for the same property.

Within the text format, the value of a property may also be retrieved by its name with `\RpgGetKey` (page 26). Using the property's variable directly works in both the text and card formats.

Formats

The format of a rule environment is a code setter (see the sidebar (page 28)). Both formats have the same arguments:

Argument	Value
#1	The title of the element.
#2	The body of the environment.

The values of the properties are available through their variables (or `\RpgGetKey`).

`\RpgTheme[X]CardFormat` { `code` }

Stores the code which typesets the element in card mode. It is typeset inside an `RpgCard` (page 25), and so continues onto further cards if it is too long for one. By default, the title is typeset as a `\subsection`, followed by the body.

`\RpgTheme[X]Format { code }`

Stores the code which typesets the element in text mode. By default, the title is typeset as a `\subsection`, followed by the body.

Key-Value Options

Themes often need to give their own commands and environments key-value options. The following commands define such options in a single line each, creating the variable in which the value is stored if it does not already exist. Each key belongs to a **family**, such as `rpg/table`, and is set with `\keys_set:nn{ family }{ key-value-list }`.

`\RpgThemeKeys { family }{ definitions }`

Defines several keys in the same **family** at once. Within the **definitions**, each of the commands below may be used without the `New` in its name and without its **family** argument (for example, `\RpgThemeTLKey{ key }{ variable }{ default }`). These short forms are only available within `\RpgThemeKeys`.

```
\RpgThemeKeys{rpg/my-env}
{
  \RpgThemeTLKey {title} {\l_myenv_title_tl} {Untitled}
  \RpgThemeBoolKey {hidden} {\l_myenv_hidden_bool}
  \RpgThemeTLChoiceKey {size} {\l_myenv_size_tl} {small,large} {small}
}

\keys_set:nn {rpg/my-env} {title=Hello, hidden, size=large}
```

`\RpgThemeNewBoolKey (*){ family }{ key }{ variable }`

`\RpgThemeBoolKey` Defines a flag, which takes no value. The unstarred form sets the boolean **variable** to false, and makes it true when the flag is present. The starred form does the opposite: the variable starts true, and the flag makes it false.

`\RpgThemeNewCodeMapKey { family }{ key }{ choices }{ code }{ default }`

`\RpgThemeCodeMapKey` Defines a key whose value must be one of the comma-separated **choices**. Rather than storing the value, it runs the corresponding entry of the comma-separated **code** list. Any entry of **code** which itself contains a comma must be enclosed in braces.

`\RpgThemeNewTLChoiceKey { family }{ key }{ variable }{ choices }{ default }`

`\RpgThemeTLChoiceKey` Defines a key whose value must be one of the comma-separated **choices**, and stores it in **variable**.

`\RpgThemeNewTLKey { family }{ key }{ variable }{ default }`

`\RpgThemeTLKey` Defines a key which takes any value, and stores it in **variable**. A value must be given whenever the key is used.

`\RpgThemeNewTLMapKey { family }{ key }{ variable }{ choices }{ values }{ default }`

`\RpgThemeTLMapKey` Defines a key whose value must be one of the comma-separated **choices**, and stores the corresponding entry of the comma-separated **values** in **variable**. For example, the `columns` option maps the choices `1,2` to the values `onecolumn,twocolumn`.

PART III

Appendices

Appendix A: The rpgl_{at}ex Compiler

rpgl_{at}ex is a Python 3 script (`scripts/rpglatex`) which wraps `xelatex` or `luaatex`, adding several conveniences for compiling R_PG_TE_X documents. To use it from any directory, place it on your `PATH`, for example with a symbolic link:

```
bash
```

```
ln -s ~/path/to/rpgtex/scripts/rpglatex ~/.local/bin/rpglatex
```

`rpglatex` [`options`] `<file>`

Compiles `<file>`, performing the following steps automatically:

- Defines `\RpgPackagePath`, so that R_PG_TE_X does not need to locate itself.
- Runs the compiler twice, so that references and `tikz remember picture` commands resolve correctly (three times with `-full`).
- Runs `bibtex` between the passes, if requested with `-ref`, and `makeindex` whenever the document produces an index.
- Keeps the auxiliary files (`.aux`, `.log` and so on) in a separate build directory, and moves only the finished PDF to the output location.
- Hides the compiler's lengthy output, printing a short summary instead. If compilation fails, the compiler's output is printed so that the error can be found.

The following options are available:

Option	Effect	Default
<code>-b, -build <dir></code>	The directory in which auxiliary files are stored.	<code>.build</code>
<code>-f, -full</code>	Runs the compiler three times, rather than twice, for very complex or reference-heavy documents.	off
<code>-l, -lua_{at}ex</code>	Compiles with <code>lua_{at}ex</code> .	off
<code>-o, -output <name></code>	The name of the finished PDF.	<code><file>.pdf</code>
<code>-p, -print</code>	Compiles in print mode, by setting <code>bg=print</code> through the command line interface (page 11).	off
<code>-q, -quiet</code>	Suppresses the script's own progress messages.	off
<code>-r, -ref</code>	Runs <code>bibtex</code> after the first pass.	off
<code>-show <0/1></code>	Whether to open the finished PDF (or switch to it, if already open) in a viewer.	1
<code>-v, -verbose</code>	Shows the compiler's full output.	off
<code>-x, -xel_{at}ex</code>	Compiles with <code>xel_{at}ex</code> .	on

For example, the following compiles a print version of `adventure.tex` with `luaatex`, without opening it:

```
rpglatex adventure.tex -l -p -show 0
```

Appendix B: Dependencies

RPGTeX loads the packages listed below. All of them are included in a full TeX Live installation, and MiKTeX installs missing packages automatically the first time they are needed. On a minimal installation, missing packages may be added with `tlmgr`:

```
bash
```

```
tlmgr install <package>
```

Some files are distributed as part of a larger package, whose name must be given to `tlmgr` instead; this name is shown in brackets. Packages marked with an asterisk are only loaded in some circumstances, which are described beneath the list.

<code>adjustbox</code>	<code>hang</code>	<code>stfloats (sttools)</code>
<code>array (tools)</code>	<code>l3keys2e (l3packages)</code>	<code>supertabular</code>
<code>contour</code>	<code>lettrine</code>	<code>tabularx (tools)</code>
<code>cuted (sttools)</code>	<code>luacolor*</code>	<code>tcolorbox</code>
<code>enumitem</code>	<code>magaz</code>	<code>tikz (pgf)</code>
<code>eso-pic</code>	<code>microtype</code>	<code>titlesec</code>
<code>fancyhdr</code>	<code>multitoc* (ms)</code>	<code>titletoc (titlesec)</code>
<code>float</code>	<code>nameref (hyperref)</code>	<code>tocloft</code>
<code>fontenc* (latex)</code>	<code>ragged2e</code>	<code>transparent</code>
<code>fontspec*</code>	<code>relsize</code>	<code>xcolor</code>
<code>geometry*</code>	<code>Royal (cfr-initials)</code>	<code>xltabular</code>
<code>graphicx (graphics)</code>	<code>standalone*</code>	<code>xparse (l3packages)</code>

The packages marked with an asterisk are loaded as follows:

- `fontenc` and `fontspec`: when the `font` option is active (page 8).
- `geometry`: by the RPGTeX classes.
- `luacolor`: when compiling with `lualatex`.
- `multitoc`: by the RPGTeX classes, unless the `nomultitoc` option is used.
- `standalone`: by the `rpgcard` and `rpgdeck` classes.

In addition, the classes are built on the standard `book` and `article` classes, which are part of every L^AT_EX installation, as is `expl3`.

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